Aradhya Pasikanti

Aspiring Robotics Engineer & Entrepreneur

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High school student and FBLA National Champion pursuing robotics engineering and business leadership, with international DECA awards and multiple VEX Robotics World Championship appearances. Experienced mentoring robotics teams, building AI/CAD driven engineering projects, and leading student organizations focusing on STEM outreach.

EDUCATION

Downingtown High School East Expected June 2028

- Class Rank: 2/416
- GPA: 4.0/4.0 unweighted, 4.7/4.0 weighted
- AP Courses completed: Physics 1 (5), Microeconomics (5), Computer Science Principles (5), World History (5), Macroeconomics (4), US History (4)
- AP Courses in progress: Physics C: Mechanics, Calculus AB, Statistics, Business & Finance, Government, Language & Comp.

ACCOMPLISHMENTS

- National Champion, Introduction to Business Presentation | FBLA: placed 1st out of 300 national competitors among 16,000+ students, delivering a presentation on Smart Money Management for High School Students
- International Award Winner, Principles of Entrepreneurship | DECA: achieved the highest roleplay (presentation) score out of 200+ international competitors among 25,000+ students in entrepreneurship and business management
- VEX Robotics Awards | V5RC: competed for 5 years, earning 26 awards across international and statewide competitions (with 120+ competitors @ states), including Judges, Innovate, Design, Skills Champion, Tournament Champion, Excellence, and multiple EPA States honors | 2x World Championship Competitor
- International Finalist, Mental Health Promotion | HOSA: top 25 entry selected out of 600+ international competitors among 15,000+ students presenting on teenage suicide prevention and mental health
- National Qualifier (Thermo-Fisher), Science Fair | DVSF: placed 3rd among 130+ tri-state competitors, analyzing the impact of biodegradable hydrogels on agricultural applications and soil hydration
- 4C Award | DASD: recognized out of 500 students within the district for exemplifying Critical Thinking, Communication, Collaboration, and Creativity

EXPERIENCE

Robotics Mentor | *VEXMEN Robotics* Apr 2023 – Present

- Provided 90+ hours of engineering mentorship for both High School and Middle School robotics teams, improving technical knowledge and competition readiness
- Led 7 teams to States and 2 teams to the World Championship over the past year through consistent mentorship on mechanical systems, programming, and design documentation

Team Leader, Team 81Z | *VEXMEN Robotics* Aug 2021 – Present

- Led over 12 unique individuals over the span of 5 years in advanced robotics engineering winning over 26 awards with 2 World Championship appearances (1/400 teams out of 17000+ teams worldwide)
- Developed expert level proficiency in Computer Aided Design, autonomous path planning, odometry, and PID control systems, along with thorough design documentation and interview skills

Co-Founder | *Uplift — Teenage Suicide Prevention Initiative* Nov 2025 – Present

- Grew an online community of 305+ members for teens struggling with mental health, creating informational, interactive, and community based campaigns that reached 40K+ impressions
- Consistently met with a Mental Health Consultant from the CHESCO Department for Suicide Prevention to ensure the project was governed safely and ethically
- Created a mental health website and podcast equipped with games, breathing exercises, teen stories, and a dynamic risk detecting AI assistant

VP of AI Integration & Development | *Vincere USA* Jun 2025 – Dec 2025

- Led the implementation and integration of AI to serve as an educational tool for foreign students lacking proper access to education

- Used AI to develop fundraising and outreach strategies, growing 2 new school partnerships in Portugal and India, impacting 60+ students and raising close to $800

President, Math Tutor Club | *Downingtown Area School District* Sep 2024 – Present

- Grew and manage a team of 40+ Math Tutors serving a student body of 1,800+
- Oversaw club organization and logistics, including peer to peer matching, administration communication, and club growth. Tutored 20+ hours of advanced math for middle and high school students

Social Media Manager | *VEXMEN Robotics* Aug 2025 – Present

- Grew Instagram and Facebook to 1100+ followers with an average of 25K views/month through educational engineering content
- Grew a YouTube channel to 170 subscribers and 45.6K total views across 13 videos

Camp & Workshop Organizer | *VEXMEN Robotics* Jul 2024 – Aug 2025

- Directed and executed 3, 5 day robotics camps, investing 90+ hours educating grades 3–9 through the full engineering design process
- Directed a 2 day Girl Scout workshop and a 5th grade STEM assembly on the real world applications of robotics

Treasurer | *Downingtown Area School District* Aug 2024 – Jun 2025

- Organized 5+ district-wide fundraisers, raising close to $1,200 in 10 months for the Downingtown East Class of 2028

Initiative Organizer | *Yale Emerging Climate Leaders Fellowship* Aug 2024 – May 2025

- Planned and executed 2 district-wide textbook drives, repurposing test prep books and redistributing them to students with limited access to educational resources

Content Creator | *cooking_.up* Dec 2025 – Present

- Grew a content account tying in depth engineering concepts to real life situations through food, reaching ~600 followers and 30K+ impressions

SKILLS

- AI/ML: Python, CNNs, generative AI, LLMs, NLP, computer vision (YOLO, sliding-window detection, gradient weighting)
- Engineering: CAD (Onshape, Fusion), robotic build/design/documentation/programming, C++, HTML
- Business & Finance: infrastructure proposal development, financial literacy analysis, investor presentations, business presentation
- Leadership: peer to peer education, team management, cross functional collaboration

INTERNSHIP & RESEARCH EXPERIENCE

AI/ML in Drug Discovery Intern | *Merck* Jul 2026 – Sep 2026

- Worked at the intersection of AI and computational immunology, applying machine learning models to vaccine design
- Focused on neoantigen testing, using AI models to evaluate candidate neoantigens for immunogenicity and their ability to activate T-cell responses
- Contributed to the design of AI-informed, T-cell activating vaccine candidates as part of Merck's drug discovery workflow

Finance & Infrastructure Intern | *Georgetown University* Jul 2026 – Aug 2026

- Worked directly under Georgetown University professors to build an infrastructure investment proposal for a startup based in Bauchi, Nigeria
- Conducted financial and infrastructure analysis to structure a viable proposal for investor consideration
- Presented proposal to a panel of investors, including a senior Starbucks executive, the Dean of Georgetown University, and other high level investors
- Proposal was approved as #1 out of 8 other teams receiving $4,000 of funding for the proposed infrastructure in the Bauchi based startup

AI Intern | *Inspirit AI* Jun 2025 – Aug 2025

- Studied AI architectures including convolutional neural networks (CNNs), generative AI, large language models (LLMs), and NLP modeling techniques (with BERT)
- Applied this foundation to a research project on object detection for self driving cars
- Developed a custom detection algorithm combining YOLO, a sliding window search approach, gradient weighting, and other image detection techniques